Next-Day Municipal Service-Demand Forecasting Across Four Cities:
A Benchmark with a Controlled Simulated Capacity-Allocation Evaluation

Nassar Naif Alsharif
Independent Researcher NassarAlsharif0@gmail.com

June 2026

Abstract

City agencies use short-horizon forecasts of 311 service-request volume as a planning input, but forecast accuracy is only a proxy for the decisions a forecast supports. We build a reproducible benchmark from 36.8 million acquired service-request records (30.2 million retained after documented exclusions) across four U.S. cities (New York, Chicago, San Francisco, Austin; 2020–2025), harmonized into documented service families and joined with NOAA station weather, and we evaluate next-day forecasters jointly with a transparent, explicitly simulated family-level capacity-allocation model under pre-specified hypothetical service-pressure regimes, with capacity budgets frozen from training data only and all model selection performed on validation data only. Calendar augmentation reduces next-day error in every city (8.6–14.0%, all 28-day block-bootstrap intervals excluding zero); target-day weather helps three cities and adds nothing in the fourth, a retained null. Pooled cross-city training, temporally censored so no training row is contemporaneous with any evaluation window, raises error significantly in three cities; zero-shot transfer degrades error by up to 149%. In the simulation, a simple proportional allocation rule dominates an exact expected-value optimization family in 11 of 12 city-regime settings under a neutral equal-weights objective — the mechanism being that point forecasts give surplus capacity no informative gradient — while, within the optimization family, using the full predictive distribution of a single quantile model significantly beats its own median and implied-mean representations in all 12 settings. Forecast quality transfers to simulated decision value within a fixed policy in all 12 settings, and validation-time selection by decision loss mostly coincides with selection by accuracy (9 of 12), occasionally gaining and never harming. All data are public, every number regenerates from one command under a guard suite that enforces the protocol, and the simulation makes no claim about any city's actual operations.

1 Introduction

Municipal 311 systems receive millions of service requests each year — heating failures, water leaks, street damage, sanitation problems, noise — and agencies plan finite daily service capacity against this demand. Short-horizon forecasts of request volume are a natural planning input, and a growing literature shows that public signals such as weather and calendar structure improve such forecasts. But planners do not consume mean absolute error; they consume allocations, and the mapping from predictive accuracy to allocation quality is neither linear nor guaranteed.

This paper contributes, first, a reproducible four-city benchmark of next-day reported 311 demand — New York, Chicago, San Francisco, and Austin, 2020–2025, built entirely from official open-data APIs with versioned harmonization rules and provenance manifests — and, second, a controlled, explicitly simulated evaluation of what forecast differences are worth to a family-level capacity allocation under pre-specified hypothetical service-pressure regimes. The governing question is:

> Across heterogeneous municipal 311 reporting systems, under pre-specified hypothetical service-pressure regimes, when do differences in next-day forecasts of reported administrative demand produce materially different outcomes in a transparent simulated family-level capacity allocation, and when are predictive rankings sufficient proxies for simulated decision rankings?

Four design commitments distinguish the evaluation protocol. **(i) Frozen environments:** capacity budgets derive from training data only and are identical at model-selection time and at the single test evaluation. **(ii) Validation-only selection:** no headline configuration is chosen using test outcomes. **(iii) Temporal censoring:** pooled and transfer models never train on rows contemporaneous with any evaluation window. **(iv) Controlled uncertainty contrast:** the value of distributional information is measured within one fitted quantile model — its median, implied-mean, and full-distribution representations — holding everything else fixed. A guard suite of automated tests enforces each commitment and fails the build if any is violated.

Throughout, we are explicit about what the data can and cannot support. Reported requests are not social need (Kontokosta and Hong, 2021); the allocation layer is a simulation over abstract request-equivalent capacity units whose parameters are varied over a pre-specified grid rather than asserted; and no causal, deployment, or operational-guidance claim is made. The study is an empirical bridge between the forecasting literature, which stops at accuracy, and the decision-focused learning literature (Elmachtoub and Grigas, 2022; Mandi et al., 2024), which rarely touches public-sector data.

2 Related Work

Public signals in operational forecasting. Cui et al. (2018) established, in retail, that publicly available signals (social media) measurably improve daily operational forecasts; Steinker et al. (2017) did the same for weather in e-commerce fulfilment. An earlier single-city study by the author brought this template to municipal service demand on NYC 311. The present paper tests whether that finding is a property of New York or of 311 systems generally, and then asks what the gains are worth inside a controlled simulated allocation.

311 analytics. 311 data underpin a largely single-city, prediction-centric literature: Cheng et al. (2022) predict request volumes and their correlates in Miami-Dade County; Kontokosta et al. (2017) and Kontokosta and Hong (2021) document socio-spatial disparities in the propensity to report, establishing that 311 volumes measure *reported* demand rather than need. We inherit the latter caveat explicitly. To our knowledge no prior work evaluates 311 forecasting across multiple cities under one protocol, nor couples it to an explicit allocation loss with frozen pre-test environments.

From prediction to decisions. That predictive accuracy and decision quality are distinct objectives is the founding observation of prescriptive analytics (Bertsimas and Kallus, 2020) and decision-focused learning: the SPO framework trains models against the downstream objective (Elmachtoub and Grigas, 2022), task-based end-to-end learning does so in stochastic programs (Donti et al., 2017), and Mandi et al. (2024) survey and benchmark the field. Those benchmarks are dominated by synthetic costs or energy/inventory/routing settings. We contribute complementary public-sector evidence: rather than proposing a training method, we measure empirically where accuracy-optimized models, distributional information, and decision-aware validation selection do and do not change simulated allocation outcomes.

Capacity planning under time-varying demand. Setting service capacity against predictable daily variation is a classical operations problem (Green et al., 2007); our simulation is a deliberately transparent discrete-time instance with integer capacity units, family-level objectives, and carryover of unresolved request-equivalents, designed for auditability rather than queueing fidelity.

Cross-city learning and urban computing. Urban computing treats the city as a sensing-and-decision platform (Zheng et al., 2014); cross-city transfer has been studied mainly for traffic and crowd-flow prediction (Wang et al., 2019), and pooled training across related series is standard in modern forecasting (Salinas et al., 2020). We test pooled-versus-local training and zero-shot transfer under explicit temporal censoring, and carry the comparison through to simulated decision quality.

3 Problem Formulation

Prediction task. Let $y_{c,s,t}$ denote the number of 311 requests created in city c and harmonized service family $s \in \mathcal{S}_c$ on calendar day t, where $\mathcal{S}_c$ is the city's *active* family set (eight families in New York, San Francisco, and Austin; seven in Chicago, where residential noise is structurally absent from the intake taxonomy). At the end of day t (the information cutoff), a forecaster produces a point forecast $\hat{y}_{c,s,t+1}$ and, for the probabilistic model, quantile forecasts $\hat{q}^{\tau}_{c,s,t+1}$, $\tau \in \{.05, .25, .50, .75, .95\}$. Endogenous features use observations through day t only; calendar features of day $t+1$ are deterministic; target-day weather uses the realized observation for $t+1$ as a stand-in for a day-ahead weather forecast (assumption A3, bounded by a lagged-weather variant).

Simulated allocation. Each evening a simulated planner allocates a fixed budget of B_c abstract capacity units across active families, $x_{c,s,t+1} \in \mathbb{Z}_{\geq 0}$, $\sum_s x_{c,s,t+1} = B_c$; a unit resolves up to κ request-equivalents the next day. Unresolved workload carries over:

$$\begin{aligned} w_{c,s,t+1} &= b_{c,s,t} + y_{c,s,t+1}, \qquad \text{resolved} = \min(w_{c,s,t+1},\, \kappa\, x_{c,s,t+1}), \\ u_{c,s,t+1} &= w_{c,s,t+1} - \text{resolved}, \qquad b_{c,s,t+1} = (1-\alpha)\, u_{c,s,t+1}, \end{aligned} \tag{1}$$

where b is simulated unresolved request-equivalent carryover (observable state at decision time) and α an abandonment rate ($\alpha = 0$ primary). The decision loss over the test horizon $\mathcal{T}$ is

$$L_c \;=\; \sum_{t \in \mathcal{T}} \sum_{s \in \mathcal{S}_c} \omega_s\, u_{c,s,t}, \tag{2}$$

with EQUAL weights $\omega_s = 1$ as the primary, neutral objective (one illustrative normative-priority scenario is a labeled sensitivity). Hypothetical service-pressure regimes set κB_c to 70%, 90%, and 110% of the city's TRAINING-window mean daily demand; the resulting integer budgets are frozen before any model selection and used unchanged for validation-stage selection and the single test evaluation. Nothing in this construction estimates any city's actual capacity, productivity, or queues.

Policies. All policies receive the same observable state and allocate the full budget: *uniform* (equal units; forecast-free floor); *proportional* (largest-remainder apportionment of $\hat{y}_{c,s,t+1} + b_{c,s,t}$); and *greedy expected-value*, which exactly maximizes $\sum_s \omega_s\, \mathbb{E}[\min(D_s, \kappa x_s)]$ over integer allocations, where D_s is the forecast demand-plus-carryover distribution — degenerate for point forecasts, piecewise-linear from the quantile model. Greedy is optimal for this objective because $\mathbb{E}[\min(D, c)]$ is concave in c. A hindsight myopic reference (greedy on realized demand) appears in supplementary diagnostics only; being per-day myopic it is not a horizon-optimal bound and is never used as a denominator or target.

Table 1: Dataset summary after preprocessing (study window 2020-01-01 to 2025-12-31; 8 harmonized service families per city; no synthetic data).

City	Requests kept	Native cat.	Panel rows	Days	Mean/day
Austin	1,487,588	346	17,536	2192	679
Chicago	4,683,510	107	15,344	2192	2,137
New York	19,669,371	271	17,536	2192	8,973
San Francisco	4,374,676	136	17,536	2192	1,996

Controlled uncertainty contrast. The value of distributional information is isolated within ONE fitted quantile model: a degenerate-median arm, an implied-mean arm, and a full-distribution arm differ only in how the same predictive distribution enters the greedy policy. The separately trained point models serve as forecasting benchmarks, not as the uncertainty comparator.

Evaluation questions. For each city and regime we compare forecast accuracy (MAE, with paired 28-day moving-block bootstrap intervals; pinball loss and coverage for quantiles) and simulated decision loss (2) (raw losses, absolute paired daily differences with block-bootstrap intervals, and percentage reduction relative to the uniform floor), asking when predictive and decision orderings agree, whether distributional information changes outcomes, and whether validation-time selection by decision loss differs from selection by accuracy.

4 Data

311 service requests. We use the official open-data portals of four U.S. cities, selected by pre-stated criteria (API uniformity, full 2019–2025 coverage, category granularity, and heterogeneity in size and climate; see the feasibility audit in the companion repository): New York (`erm2-nwe9`, NYC Open Data), Chicago (`v6vf-nfxy`, current 311 system, live since December 2018), San Francisco (`vw6y-z8j6`, DataSF, PDDL license), and Austin (`i26j-ai4z`). Acquisition uses server-side daily aggregation (counts per day $\times$ native category) through the SODA API; every query URL, retrieval timestamp, and SHA-256 checksum is recorded in versioned manifests. Chicago rows flagged as duplicates by the source system are removed (523,055 requests), as are documented non-service categories: Chicago's "311 INFORMATION ONLY CALL" log entries (4,184,158 records) and aviation-routed aircraft-noise filings (1,918,574 records), and San Francisco's administrative duplicate markers. The study window is 2020-01-01 through 2025-12-31, the full scope of NYC's current dataset. As an acquisition-integrity check, the NYC layer was produced by two independent methods — server-side aggregation and client-side aggregation of a streamed export — which agreed exactly (270,978 aggregated rows).

Harmonized service families. Each city's native taxonomy ($\sim$one to three hundred categories) is mapped to eight service families — noise; housing/building; water/sewer; public safety; streets/transport; sanitation; parks/trees; other — by an ordered, case-insensitive, versioned keyword rule set applied identically to all cities. The rules were audited against each city's real category inventory and frozen before any model was trained; after the freeze, the unmapped "other" share is 6.2% (New York), 6.9% (Chicago), 8.8% (San Francisco), and 23.5% (Austin, whose intake includes large animal-services and explicitly generic categories that fit no family and are deliberately retained in the served "other" bucket). Families are *operationally analogous planning buckets*, not identical workloads: Chicago has no noise family at all (residential noise is not a Chicago 311 request type), the supplement reports the full native-category composition of every family in every city, and no cross-city claim in this paper assumes compositional identity.

Weather. NOAA GHCN-Daily station observations (precipitation, max/min temperature, snowfall, snow depth; derived mean temperature) from one first-order station per city (Central Park; O'Hare; San Francisco Downtown; Camp Mabry), retrieved from NOAA's AWS Open Data mirror with checksums. Quality-flagged values are dropped; short temperature gaps (≤ 7 days) are interpolated; missing snow fields are treated as zero. Wind speed was excluded for all cities because the San Francisco station does not report it and Central Park has a multi-month gap — a documented deviation from the single-city study this work extends. A single station per city is a documented spatial simplification (A2). No synthetic demand or synthetic weather is used anywhere in the study.

5 Methods

Feature sets. Four nested feature sets isolate the marginal value of each public signal: *internal* (lags $1, 2, 3, 7, 14, 28$ and 7/28-day rolling mean and standard deviation, windows ending at day t); *calendar* (internal + deterministic day-of-week, month, weekend, U.S. federal holiday, and smooth day-of-year terms for the target day); *weather* (internal + target-day and current-day station weather); and *calendar+weather*. Family indicators are one-hot encoded; pooled models add city indicators.

Models. Two forecast-free baselines (trailing 7-day mean; same-weekday seasonal naive), ridge regression, a random forest (Breiman, 2001), and LightGBM with an L1 objective (Ke et al., 2017) form the pre-specified point-model grid; quantile-objective LightGBM (following Koenker and Bassett, 1978) provides the predictive distribution, with non-crossing enforced by cumulative maxima. Hyperparameters are fixed across cities and feature sets; baselines are never dropped from tables.

Protocol. Within each city, the earliest 70% of days form training, the next 15% validation, and the final 15% an untouched test window. For every (city, feature set), the model is selected on VALIDATION MAE, frozen, refit on train+validation, and evaluated once on test; exhaustive test grids appear only as supplementary descriptive evidence. The quantile model is fit on training data alone for all validation-stage outputs and refit on train+validation for its single test evaluation. Stability is assessed with five expanding-window rolling-origin folds (Hyndman and Athanasopoulos, 2021). Pooled ("global") models train on all four cities with stage-specific temporal censoring: validation-stage fits use only rows dated no later than the earliest training cutoff across cities, test-stage fits no later than the earliest validation cutoff, so no pooled training row is contemporaneous with any evaluation window. Leave-one-city-out transfer is likewise censored at the held-out city's validation cutoff. All accuracy contrasts use paired moving-block bootstrap intervals on daily mean absolute-error differentials (28-day blocks, 2,000 resamples, seed 20260609; 14- and 56-day blocks as pre-specified sensitivity) (in the spirit of Diebold and Mariano, 1995).

Decision evaluation. Every point forecaster drives the proportional and greedy policies; the quantile model drives the three uncertainty arms; uniform allocation is the forecast-free floor. Simulations run day-by-day over each city's test window at the frozen budgets with $\kappa = 50$ and equal weights. Pre-specified sensitivities: one normative-priority scenario; one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family; 10% abandonment; train+validation budget calibration; and an Austin variant that excludes the heterogeneous *other* family with budgets recomputed accordingly. Validation-time model selection by decision loss (simulated on the validation window at the same frozen budgets) is compared against selection by validation MAE, both frozen before test.

Guard suite. Thirteen automated guards enforce the protocol and fail the build otherwise, including: budgets reproducible from training data and invariant to test perturbation; selection invariant to test metrics; recomputed censoring dates on every transfer row and a fit-time proof manifest for pooled censoring; identical fitted model behind all uncertainty arms; structural absence never encoded as zero; exact budget conservation; forbidden-terminology scans of all manuscript-facing artifacts; configuration–output agreement for every sensitivity scenario; source-identifier consistency; and provenance hashes binding every table and figure to the generating run.

6 Forecasting Results

All headline aggregations use validation-only selection; exhaustive grids are supplementary. Intervals are paired 28-day moving-block bootstrap (stable at 14 and 56 days throughout).

Calendar gains replicate everywhere. The validation-selected calendar model lowers held-out test MAE relative to the validation-selected internal-history model in every city: −11.9% in New York (207.0 → 182.4), −15.6% in Chicago (60.4 → 50.9), −10.4% in San Francisco (35.2 → 31.5), and −10.2% in Austin (16.5 → 14.8); the fixed-estimator (LightGBM) calendar effect is significant in all four cities. Against the same-weekday seasonal-naive baseline, the selected calendar+weather model reduces test MAE by 41.4% (New York), 41.5% (Chicago), 31.3% (San Francisco), and 31.3% (Austin). Validation-only selection matters in its own right: on the internal feature set it picks the random forest in Austin and Chicago and ridge in San Francisco, choices a test-minimum rule would have hidden.

Weather's marginal value is real but city-dependent. On top of calendar features, target-day weather further reduces MAE by 9.3% in New York, 3.3% in Chicago, and 4.7% in San Francisco (all intervals excluding zero) — and by exactly nothing in Austin (+0.001 MAE; interval covers zero). The Austin null is retained as a finding. The lagged-weather variant (assumption A3 removed) erases most of New York's weather gain (178.6 versus 165.4, against a 182.4 calendar-only reference), so much of the weather value is conditional on an accurate day-ahead weather forecast — a realistic but explicitly stated premise.

Stability. Five expanding-window rolling-origin folds reproduce the feature-set ordering (internal > calendar > calendar+weather in MAE) in every city; the headline comparisons are not artifacts of a single split.

Pooling and transfer mostly hurt under valid censoring. With stage-censored training (no pooled training row contemporaneous with any evaluation window), the pooled model raises test MAE relative to local models in Austin (+4.1%), Chicago (+6.5%), and New York (+1.8%), each interval excluding zero, and is neutral in San Francisco (−1.2%, interval covers zero); validation selection accordingly picks the LOCAL calendar+weather model in all four cities. Temporally censored zero-shot transfer (training without the target city, censored at its validation cutoff) degrades MAE by +5.4% (San Francisco) to +148.5% (New York). With six years of local history, cross-city pooling and transfer are not beneficial in this benchmark; the negative result is retained.

Probabilistic forecasts are sharp but under-dispersed. The quantile model's central 90% intervals cover 76.8–81.2% of test outcomes. We report this miscalibration rather than re-tuning it away; Section 7 shows the distributional information is nonetheless valuable within the optimization policy class, and conformal recalibration is noted as future work.

7 Simulated Capacity-Allocation Evaluation

All simulations run at budgets frozen from training data (New York 119/153/187 units across the scarce/moderate/generous regimes; Chicago 29/38/46; San Francisco 26/33/41; Austin 8/11/13; $\kappa = 50$), under the neutral equal-weights objective. Losses are raw simulated unresolved request-equivalents; policy contrasts carry paired 28-day block-bootstrap intervals on daily loss differentials. Four findings emerge.

A simple proportional rule dominates the exact expected-value family. Proportional allocation driven by the validation-selected forecaster attains the lowest simulated loss in 11 of 12 city-regime settings, beating even the full-distribution arm of the quantile model by 0.1–6.5% (the exception is San Francisco under generous capacity, +1.3% for the full arm); against the point-forecast greedy policy the proportional advantage is significant in all 12 settings. The mechanism is identified, not conjectured: under a degenerate point-forecast distribution, capacity beyond forecast-plus-carryover has zero marginal expected value, so the exact optimizer places surplus units without information, while proportional apportionment distributes them in proportion to expected workload. Relative to the forecast-free uniform floor, the proportional policy with the selected forecaster reduces simulated loss by 18–23% (scarce), 38–70% (moderate), and 60–93% (generous) across cities. This dominance is a property of the simulated environment and its neutral objective — under the normative-priority sensitivity the gap compresses — and it constitutes no guidance about any city's actual operations.

Within the optimization family, the full distribution beats its own point representations everywhere. Holding the single fitted quantile model and every other input fixed, the full-distribution arm significantly outperforms both the median arm and the implied-mean arm in 12 of 12 settings. Distributional information breaks the surplus-placement ties that disable the point-forecast optimizer — evidence that *how* a forecast enters a policy can matter as much as forecast accuracy itself.

Forecast quality transfers to simulated decision value within a fixed policy. Replacing the trailing-mean baseline with the validation-selected forecaster, policy held fixed, reduces simulated daily loss significantly in all 12 settings (mean daily reductions from ≈1.8 thousand request-equivalents in Austin to ≈72 thousand in New York under scarcity).

Accuracy-based and decision-based selection mostly agree. Selecting the deployment forecaster by validation decision loss instead of validation MAE — both under the frozen budgets — yields the same choice in 9 of 12 settings, a significant gain in 2 (Austin, scarce and moderate, where the decision criterion prefers the local over the pooled model), and one interval-tie; it never harmed.

Decision-based validation selection is therefore a cheap, safe check rather than an established improvement.

8 Robustness and Sensitivity

Inference stability. None of the 48 decision-contrast conclusions changes when the bootstrap block length moves from the primary 28 days to 14 or 56 days; the same holds for the forecast contrasts.

Decision-layer sensitivity. The pre-specified grid preserves every qualitative conclusion: with 10% daily abandonment, policy ordering is unchanged in all cities; with one-family-at-a-time service-yield multipliers of 0.70 and 1.30 over every active family, the ordering direction never reverses (the largest perturbation is New York, generous regime, noise yield $\times 1.30$); with budgets calibrated on train+validation instead of training only, conclusions are unchanged; and the Austin analysis with the heterogeneous *other* family excluded (budgets recomputed) produces the identical policy ordering as the primary analysis, so Austin's conclusions do not depend on its residual category. Under the illustrative normative-priority scenario the proportional–optimizer gap compresses, which is why the proportional-dominance claim is stated as conditional on the neutral objective.

What would change the conclusions. The simulation results are conditional on the stylization assumptions (interchangeable units, fixed yield, no intra-day dynamics) and on the equal-weights objective; we claim regime *patterns* within this simulated environment, not city-specific magnitudes. The forecasting results are unconditional on those assumptions; the weather findings are conditional on day-ahead forecast availability (A3), bounded by the lagged-weather variant.

9 Responsible Use of Reported-Demand Data

311 records measure the propensity to report as much as the incidence of problems: communities differ systematically in access, trust, language, and expectation that reporting leads to action (Kontokosta et al., 2017; Kontokosta and Hong, 2021), and prediction-driven prioritization can institutionalize whatever bias the data carry (Samorani et al., 2022). Three design consequences follow in this study. First, the target variable is named throughout as *reported* demand, and no result is interpreted as a measurement of need. Second, because our spatial unit is the whole city, the most direct channel for reporting-bias amplification — reallocating resources *away* from low-reporting neighbourhoods — is outside the action space of the simulated planner; the family-level channel remains, and we report per-family unserved-demand distributions for every policy so that systematic deprioritization of any family is visible rather than hidden in aggregate scores. Third, the priority weights ω_s encode a value judgment, not a fact; we expose them as configuration, vary them over a pre-specified grid in sensitivity analysis, and regard their selection in any real deployment as a decision that belongs to accountable public officials, not to a model. A forecast-driven allocation system used in practice would additionally require visibility into where reporting data are likely incomplete — a measurement problem this paper's data cannot solve and that we therefore flag as a limitation rather than simulate.

10 Limitations

- **Explicit simulation boundary.** The allocation layer is a transparent simulation over abstract request-equivalent capacity units; it omits shifts, travel, specialization, geography, and intra-day dynamics, and none of its parameters estimates any city's actual capacity, productivity, or queues. Conclusions concern the structure of the accuracy-to-allocation mapping inside this simulated environment.

- **Objective dependence.** The proportional-dominance result is conditional on the neutral equal-weights objective; the normative-weight sensitivity compresses the gap. No weighting here is socially validated.

- **Reported demand only.** Targets and losses are functions of reported requests; unreported need is invisible to both the forecasters and the simulated planner (Section 9).

- **Harmonization is a modeling choice.** The service families are deterministic keyword buckets; native taxonomies differ across cities (Chicago lacks a residential-noise intake type; Austin's residual *other* is large, though its exclusion leaves conclusions unchanged). All mappings and compositions are published.

- **Weather information set.** Target-day weather uses realized observations as a proxy for a day-ahead forecast (A3), bounded by the lagged-weather variant; one station per city coarsens spatial variation (A2); wind was excluded for data-completeness reasons.

- **Calibration.** The quantile model's intervals are under-dispersed (90% nominal, 77–81% empirical); conformal recalibration is future work.

- **Scope.** Four U.S. Socrata-based systems; fixed a priori hyperparameters; no deep or foundation models; no causal claims; no deployment; no measurement of social need.

11 Conclusion

We built a guarded, reproducible four-city benchmark of next-day reported 311 demand and coupled it to a controlled, explicitly simulated capacity-allocation evaluation with frozen pre-test environments and validation-only selection. The corrected evidence supports a precise set of claims: public calendar signals improve next-day forecasts in every system and weather helps heterogeneously; temporally valid pooling and transfer mostly hurt; within the simulation, the choice of allocation rule can matter more than forecast refinement, a simple proportional rule dominating an exact expected-value optimizer whose point-forecast inputs leave surplus capacity uninformed; the same optimizer recovers significant value from the full predictive distribution of a single quantile model in every setting; and decision-aware validation selection mostly coincides with accuracy-based selection while never harming. The broader lesson for forecast-driven public operations research is methodological: evaluation environments must be frozen before selection, selection must never touch test outcomes, and the interface between a forecast and a policy deserves the same scrutiny as the forecast itself. Natural continuations include conformal recalibration, richer service-time modeling where closure data permit, spatial disaggregation where geographies allow honest comparison, and extension beyond U.S. systems as comparable public data emerge.

Reproducibility statement. All data are public; the companion repository contains acquisition scripts with exact query URLs, SHA-256 manifests for every raw file, deterministic seeds, unit tests (including leakage and allocation-optimality checks), and a `Makefile` that regenerates every number, table, and figure in this paper from the versioned raw data with a single command.

References

Dimitris Bertsimas and Nathan Kallus. From predictive to prescriptive analytics. *Management Science*, 66(3):1025–1044, 2020. doi: 10.1287/mnsc.2018.3253.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Shaoming Cheng, Sukumar Ganapati, Giri Narasimhan, and Farzana Beente Yusuf. A machine learning-based analysis of 311 requests in the Miami-Dade county. *Growth and Change*, 53(4): 1627–1645, 2022. doi: 10.1111/grow.12578.

Ruomeng Cui, Santiago Gallino, Antonio Moreno, and Dennis J. Zhang. The operational value of social media information. *Production and Operations Management*, 27(10):1749–1769, 2018. doi: 10.1111/poms.12707.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Priya L. Donti, Brandon Amos, and J. Zico Kolter. Task-based end-to-end model learning in stochastic optimization. In *Advances in Neural Information Processing Systems 30*, pages 5484–5494, 2017.

Adam N. Elmachtoub and Paul Grigas. Smart "predict, then optimize". *Management Science*, 68 (1):9–26, 2022. doi: 10.1287/mnsc.2020.3922.

Linda V. Green, Peter J. Kolesar, and Ward Whitt. Coping with time-varying demand when setting staffing requirements for a service system. *Production and Operations Management*, 16(1):13–39, 2007. doi: 10.1111/j.1937-5956.2007.tb00164.x.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, 3rd edition, 2021.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.

Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

Constantine E. Kontokosta and Boyeong Hong. Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. *Sustainable Cities and Society*, 64:102503, 2021. doi: 10.1016/j.scs.2020.102503.

Constantine E. Kontokosta, Boyeong Hong, and Kristi Korsberg. Equity in 311 reporting: Understanding socio-spatial differentials in the propensity to complain. *arXiv preprint arXiv:1710.02452*, 2017.

Jayanta Mandi, James Kotary, Senne Berden, Maxime Mulamba, Víctor Bucarey, Tias Guns, and Ferdinando Fioretto. Decision-focused learning: Foundations, state of the art, benchmark and future opportunities. *Journal of Artificial Intelligence Research*, 80:1623–1701, 2024. doi: 10.1613/jair.1.15320.

David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3): 1181–1191, 2020. doi: 10.1016/j.ijforecast.2019.07.001.

Michele Samorani, Shannon L. Harris, Linda Goler Blount, Haibing Lu, and Michael A. Santoro. Overbooked and overlooked: Machine learning and racial bias in medical appointment scheduling. *Manufacturing & Service Operations Management*, 24(6):2825–2842, 2022. doi: 10.1287/msom.2021.0999.

Sebastian Steinker, Kai Hoberg, and Ulrich W. Thonemann. The value of weather information for e-commerce operations. *Production and Operations Management*, 26(10):1854–1874, 2017. doi: 10.1111/poms.12721.

Leye Wang, Xu Geng, Xiaojuan Ma, Feng Liu, and Qiang Yang. Cross-city transfer learning for deep spatio-temporal prediction. *arXiv preprint arXiv:1802.00386*, 2019. Also presented at IJCAI 2019.

Yu Zheng, Licia Capra, Ouri Wolfson, and Hai Yang. Urban computing: Concepts, methodologies, and applications. *ACM Transactions on Intelligent Systems and Technology*, 5(3):38:1–38:55, 2014. doi: 10.1145/2629592.

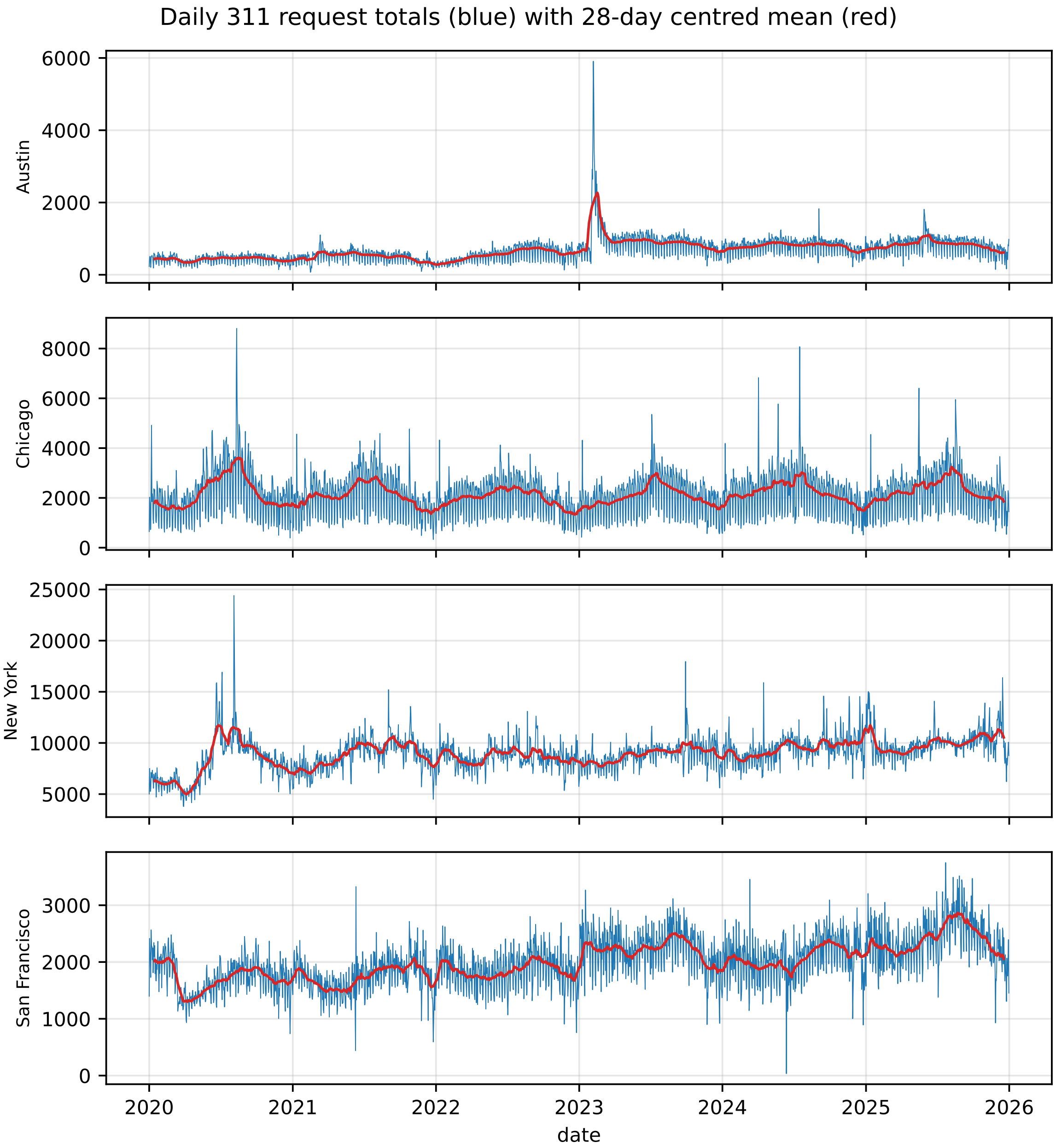

Figure 1: Daily citywide 311 request totals (blue) with 28-day centred mean (red), 2020–2025. Scale, seasonality, and disruption patterns differ visibly across the four systems.

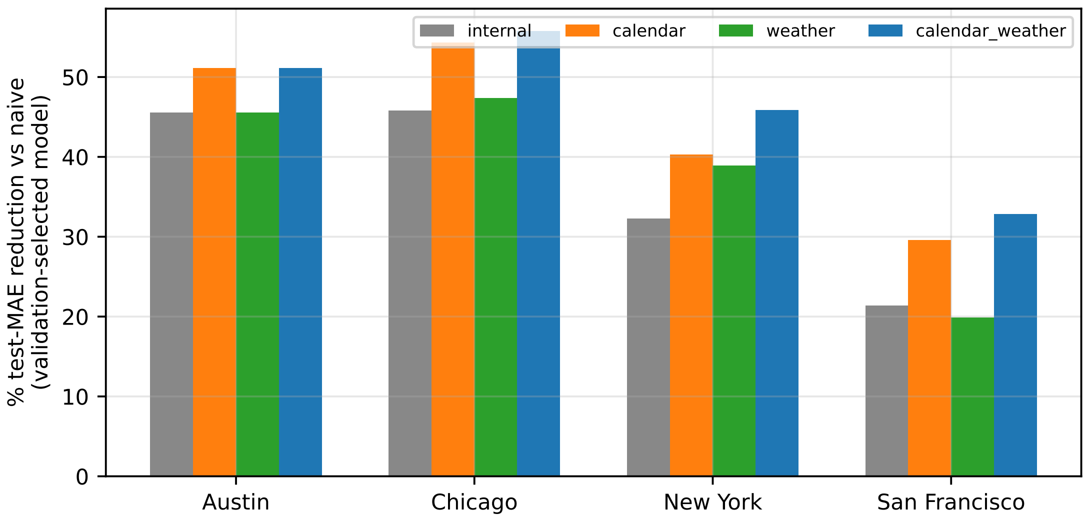

Figure 2: Held-out test-MAE reduction versus the trailing-mean baseline for the validation-selected model of each feature set and city.

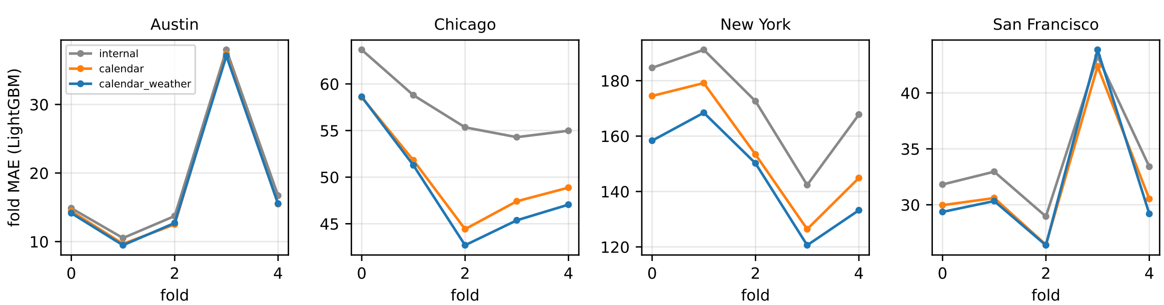

Figure 3: Rolling-origin fold MAE (LightGBM) by feature set: the calendar and weather orderings are stable across folds in all cities.

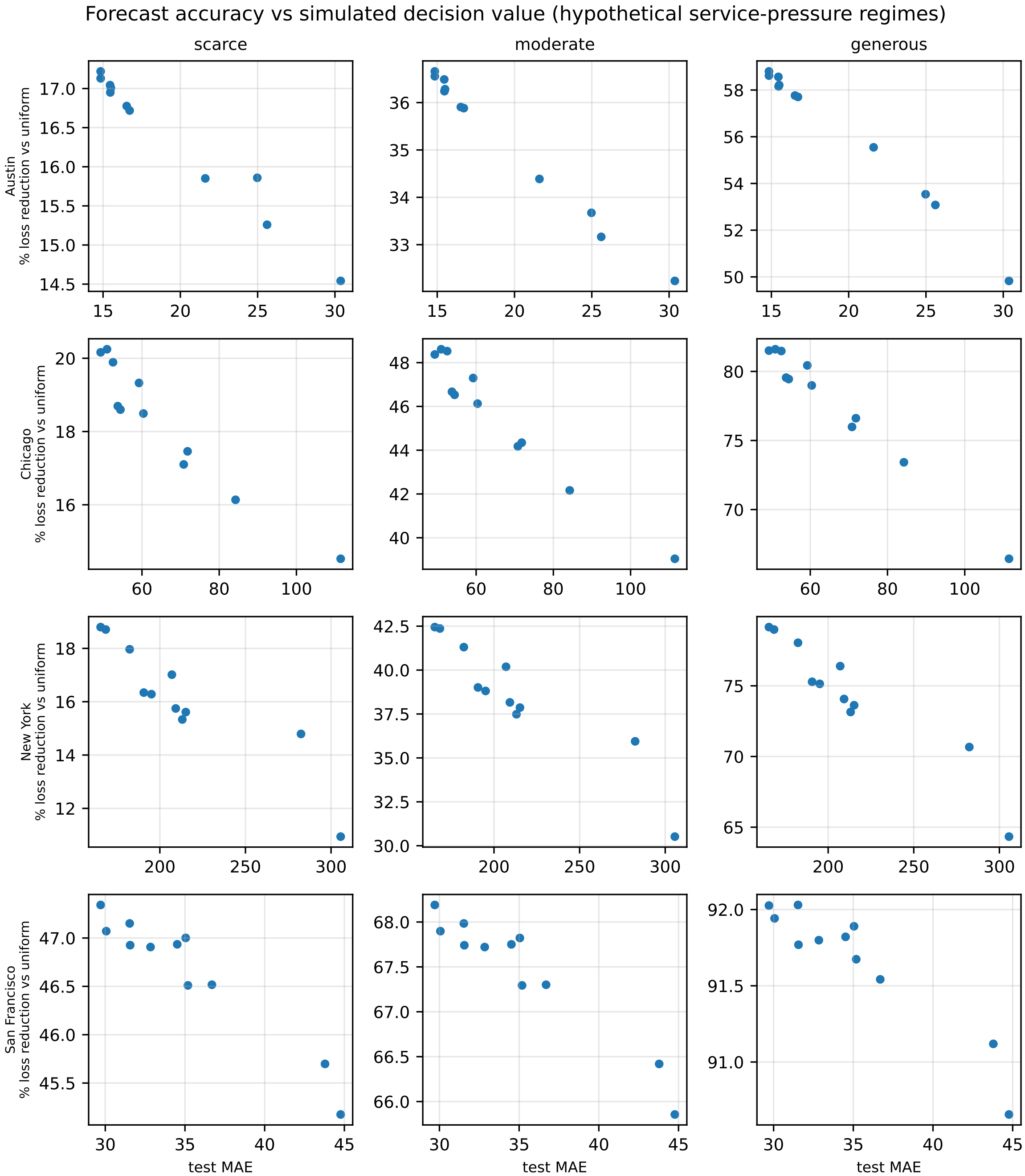

Figure 4: Forecast accuracy (test MAE, x) versus simulated decision value (% loss reduction relative to the uniform floor, y) for the pre-specified point-forecast grid under the greedy expected-value policy, by city and hypothetical service-pressure regime.